

Q1 The purpose of a loop structure is to execute a set of statements over and over again based on a condition. It performs tasks like summing a set of numbers from the user.

Q2 The difference between 'while' and 'do-while' is, in the 'do-while' statement the condition is not evaluated until after the first execution of the loop. It executes at least once.

Q3 In the mastery section, and the infinite loops part.

Q4 a) An infinite loop is a result of when the 'while' continues and the condition never becomes false.

b) 1: `int num = -1;`
 `While (num < 0) {`
 `Num = -1;`
 `}`

2: `while true;`
`System.out.println("True");`

c) Overflow occurs when there are not enough bits to store a number, it may generate a run-time error. It changes the sign of the number stored.

Q5 60 times

Q6 When x's initial value is not a number

Q7 counter records how many times a loop runs, and accumulator records the 'sum' of total and an average.

Q8 `int sum = 0;`
`for (int num = 3; num <=10; num++){`
 `sum += num;`
`}`
`System.out.println(sum);`

Q9 1. whether you know the times you want to run the code, use the 'for' statement if you know exactly how many times to loop. Otherwise use 'while-if' or 'do-while' statements.

2. Whether you know the condition should be checked before the statement. If it should, use a 'while' loop; if not use a 'do-while' loop.

Q11 a) 10

b) 'my '

c) my string.

d) MY STRING.

e) my string.